Glass ceiling unshattered: asymmetric gender culture among immigrants in different German tracks

Sunny Li*

March 23, 2026

Abstract

Talent and support go hand in hand in striving for academic improvement. How do immigrant culture and the aspirational component interact with the German tracking system? Studying the evolution of the gender gap between ages 12 and 15 with $N = 3{,}504$ students, I establish the following stylized facts: the gender gap consistently widens across different track systems; in Gymnasium, immigrant girls face a compounded disadvantage relative to non-immigrant girls; but in comprehensive schools, immigrant girls are relatively protected. I establish more causal and theoretical narratives using matched samples and structural estimation of aspiration thresholds.

1 Introduction

Immigrant families carry an aspiration premium. They consistently hold higher educational ambitions for their children than comparable native families, even after controlling for socioeconomic background (Kao and Tienda, 1995; Feliciano and Lanuza, 2017; Salikutluk, 2016). This premium is a powerful resource; immigrant boys who reach the academic track

*Department of Political Science, Stanford University. Email: `dionysus@stanford.edu`.

in Germany significantly outperform native boys. But it is not gender-neutral. This paper asks: *when does the immigrant aspiration premium become a liability for girls?*

The answer, I argue, depends on the institutional environment. In the German three-track school system, students are sorted at the end of primary school into the academic *Gymnasium*, the intermediate vocational *Realschule*, or the mixed *Gesamtschule*. Teacher recommendations heavily mediate this sorting along lines of socioeconomic background, gender, and immigration status (Dustmann et al., 2016; Jürges and Schneider, 2011). Once sorted, students are exposed to radically different peer environments, teacher expectations, and curricular demands. The aspiration literature predicts that these environments should produce divergent effort dynamics: students whose aspirations are attainable should invest effort, while those facing an aspiration gap that is too large should withdraw (Ray, 2006; Genicot and Ray, 2017; Penn, 2017). I argue that immigrant families' traditional gender norms create a *gendered aspiration threshold*, a glass ceiling that activates only when the institutional environment demands sustained, high-intensity academic commitment.

I test this hypothesis by pooling two cohorts from the German National Educational Panel Study, and estimating value-added models of Grade 9 math competence conditional on Grade 7 scores. The value-added specification is a difference-in-differences of the gender gap: by conditioning on Grade 7 achievement, the coefficient on the female indicator captures the gap that *develops* during secondary school, net of everything that came before.

The findings tell a striking story. The math gender gap widens significantly in all three tracks: girls who started at the same level as boys fall meaningfully behind over the course of secondary school, and this pattern is stable regardless of how many demographic controls are added. The crucial discovery is that immigration status moderates this gap in opposite directions depending on the track. In Gymnasium, immigrant girls face a compounded disadvantage; the high-aspiration academic environment amplifies rather than mitigates the gender gap for immigrant students. In Comprehensive schools, the pattern reverses entirely, with immigrant girls relatively protected and even slightly advantaged in the lower-pressure

environment. Socioeconomic status, by contrast, does not significantly moderate the gender gap in any track once immigration is controlled, pointing to a cultural rather than material mechanism.

I push beyond reduced-form evidence in two ways. A matched sample balanced on pre-tracking scores shows that the Gymnasium gender penalty roughly doubles relative to the pooled estimate, and the compounded immigrant-girl disadvantage strengthens in tandem. Structural estimation of Penn's (2017) piecewise threshold model across eight demographic cells reveals that Gymnasium immigrant girls are the *only* group that exhibits the aspiration trap: below the threshold, their scores lock in; above it, they actually regress. In every other cell, effort compounds above the threshold as expected. The glass ceiling is not a uniform drag. It is a regime switch at the intersection of gender, immigration, and institutional context.

These findings connect four previously separate literatures, on aspiration economics (Genicot and Ray, 2017; Dalton et al., 2016), identity and education (Akerlof and Kranton, 2002; Shayo, 2009), the immigrant paradox (Feliciano and Lanuza, 2017; Tjaden and Hunkler, 2017), and educational tracking (Hanushek and Woessmann, 2006; Van de Werfhorst and Mijs, 2010), through a single mechanism: the gendered aspiration threshold. The contribution is to show that the same cultural resource of high aspirations that benefits immigrant boys in the academic track becomes a liability for immigrant girls, not because the family lacks ambition but because the ambition is gendered in ways that bind only under high institutional pressure. The glass ceiling is unshattered not for lack of aspiration, but because aspiration itself is asymmetrically allocated.

The paper proceeds as follows. Section 2 reviews the literature on aspirations, identity, the immigrant paradox, and tracking. Section 3 develops the glass ceiling hypothesis. Sections 4 and 5 describe the data and empirical strategy. Section 6 presents the main results. Section 7 reports matched-sample and structural estimation evidence. Section 8 concludes.

2 Literature

This paper sits at the intersection of four literatures: the economics of aspirations and effort, identity and education, the immigrant aspiration paradox, and the institutional effects of educational tracking. I review these bodies of work as an integrated whole, building toward the theoretical gap that the glass ceiling hypothesis fills.

The economics of aspirations begins with the observation that effort is not monotonically increasing in ambition. Ray (2006) formalizes the key insight that aspirations are reference points, and the *gap* between current status and the aspiration threshold determines investment behavior. When the gap is moderate, individuals exert effort to close it; when the gap is too large, effort collapses because the threshold appears unreachable. Genicot and Ray (2017) embed this in a general equilibrium model where aspirations are socially determined, showing that the interaction between individual reference points and aggregate inequality produces self-reinforcing polarization. Dalton et al. (2016) extend the logic to poverty traps, demonstrating that behavioral biases in aspiration-setting compound material deprivation, so that the poor underinvest not because they are irrational but because their aspirations are endogenously compressed. Empirically, Orkin et al. (2023) demonstrate that a simple aspiration-setting workshop for poor Kenyan households raises investment and living standards, confirming that the aspiration channel is real and manipulable. Beaman et al. (2012) show that exposure to female leaders in Indian villages substantially closes the gender aspiration gap for both parents and adolescents, establishing that role models can shift the threshold.

A critical refinement comes from Penn (2017), who introduces a *class-specific* aspiration threshold that depends on the social context. In her model, individuals who fall below the class-appropriate threshold compensate for low self-esteem by distorting value identifications and withdrawing effort from their innate comparative advantages. The result is a lock-in equilibrium in which students below the threshold do not gradually improve but instead freeze. This model generates a sharp empirical prediction that distinguishes it from the

smooth aspiration-gap framework of Genicot and Ray (2017): there should be a piecewise discontinuity in the persistence of achievement, with high persistence below the threshold and low persistence above it. Section 7 tests this prediction directly.

The question of *why* students withdraw effort connects directly to the economics of identity. Akerlof and Kranton (2002) argue that identity, meaning the social categories to which individuals belong and the behavioral prescriptions attached to those categories, is a primary determinant of effort in school. Students invest in academics when doing so is consistent with their identity; they withdraw when academic effort conflicts with group norms. Shayo (2009) generalizes this in a model where individuals choose group identities to maximize a combination of material payoffs and social status. The tracked school system makes this choice particularly stark because it concentrates peers who share the same institutional trajectory, forcing students to commit to an academic identity or retreat into an alternative peer group. The "acting white" literature operationalizes this tradeoff for minority students. Austen-Smith and Fryer Jr (2005) model a signaling equilibrium in which high-ability minority students face peer rejection for investing in education, because academic success signals departure from the in-group. Roland G. Fryer and Torelli (2010), using Add Health data, find that the phenomenon is real but contextual, strongest in integrated schools where the social cost of boundary-crossing is highest. Bisin et al. (2011) show that oppositional identities can persist in dynamic equilibrium when cultural segmentation is sufficiently high, creating a mechanism for intergenerational transmission of underinvestment.

The immigrant version of this identity conflict is distinct. Immigrant children do not face peer rejection for "acting white" in the same sense. Rather, they face *family* pressure when academic commitment conflicts with gender-specific cultural expectations. Dahl et al. (2022) document this mechanism directly: granting birthright citizenship to immigrant children in Germany improves labor market integration for boys but backfires for girls, who are caught between host-country norms and parental cultural expectations. The mechanism is not peer-driven but family-driven, and it is gendered. Phalet and Schönpflug (2001) find that second-

generation daughters in Turkish-German families shift toward egalitarian values, but sons remain as conservative as their fathers. The implication is that the family's enforcement of traditional roles falls disproportionately on girls whose behavior deviates from expectations.

These identity dynamics interact with a robust empirical regularity: immigrant families hold higher educational aspirations for their children than comparable native families, even after controlling for socioeconomic background. Kao and Tienda (1995) established this "immigrant optimism" with U.S. data, showing that first- and second-generation youth outperform expectations based on their parents' human capital. Feliciano and Lanuza (2017) demonstrate that the paradox persists intergenerationally, with children of immigrants maintaining higher aspirations than third-plus generation peers. In Germany specifically, Salikutluk (2016) shows that Turkish-origin students aim higher than natives at every level of prior achievement, driven by intergenerational transmission of upward mobility goals. Pomianowicz (2024) confirms this cross-nationally and identifies educational tracking as a key moderator: the aspiration-achievement gap is largest in stratified systems where high aspirations do not automatically translate into high-track placement. The paradox has a dark side, however. Tjaden and Hunkler (2017) argue that immigrant optimism can become an "optimism trap" in tracked systems, where families choose demanding tracks at lower GPAs, increasing non-completion risk. Becker et al. (2023), using NEPS data, track how immigrant aspirations evolve during lower secondary school and find that aspirations remain high but increasingly diverge from realistic expectations. Dollmann (2021) shows that positive ethnic choice effects exist across European countries but are strongest in systems with early, binding track decisions, precisely the institutional context that amplifies the consequences of misaligned aspirations.

The cultural channel behind the math gender gap is now well established. Guiso et al. (2008), using PISA data across 40 countries, show that the gender gap in math disappears in more gender-equal cultures. Nollenberger et al. (2016) use the epidemiological approach with second-generation immigrants to isolate the cultural channel, finding that parental gender

beliefs from the country of origin account for at least two-thirds of the math gender gap in the host country. Stoet and Geary (2018) document a "gender-equality paradox" in which STEM gender gaps are *wider* in more gender-equal countries, suggesting that increased choice amplifies pre-existing preference differences when material constraints are relaxed. On the supply side, Carlana (2019) shows that implicit teacher stereotypes, measured by Implicit Association Tests, widen the math gender gap by steering girls away from STEM-congruent tracks. In Germany, Tobisch and Dresel (2017) find that teachers assign lower expectations to students with Turkish-sounding names, and Jürges and Schneider (2011) show that boys and younger students receive systematically worse Gymnasium recommendations. Carlana et al. (2022) find that a mentoring program for high-achieving immigrant students improves aspirations and reduces segregation for boys but has limited effects on girls, a pattern consistent with the gendered ceiling this paper documents.

These cultural and identity dynamics play out within an institutional structure that amplifies inequality. The German three-track system is among the most stratified in the OECD. Hanushek and Woessmann (2006) establish that early tracking increases educational inequality without apparent aggregate performance gains. Brunello and Checchi (2007) confirm that earlier tracking strengthens the link between parental SES and student performance. Van de Werfhorst and Mijs (2010) synthesize the comparative evidence, concluding that between-school streaming, as practiced in Germany, is more segregating than within-school course-by-course tracking, because it bundles peer environment, teacher quality, and curricular demand into a single institutional package. For immigrants, tracking compounds disadvantage. Ruhose and Schwerdt (2016) show that tracking produces meaningfully negative effects on immigrant test scores relative to natives across reading, math, and science. Lüdemann and Schwerdt (2013) find that second-generation immigrants receive worse grades and track recommendations than comparable natives, though socioeconomic background accounts for much of the gap. Crul and Schneider (2010) argue that institutional arrangements, not individual characteristics, are the primary determinant of second-generation outcomes,

with tracked systems being particularly consequential.

Yet the long-term effects are more ambiguous. Dustmann et al. (2016), exploiting cross-state variation in the age of tracking, find zero average long-term wage effects, suggesting that track misallocation washes out by adulthood. This null result, however, pools across demographic subgroups. The present paper argues that the aggregate null masks stark heterogeneity: tracking may have no average effect precisely because it helps some groups, such as immigrant boys in Gymnasium who benefit from the aspiration premium, while hurting others, such as immigrant girls who hit the glass ceiling. The contribution of this paper is to decompose the tracking effect along the gender-by-immigration margin and identify the aspiration mechanism that drives the divergence.

The literatures reviewed above supply the ingredients but do not combine them. The aspiration literature models effort as a function of the gap between current status and a threshold, but does not consider gendered thresholds. The identity literature models group-specific behavioral prescriptions, but treats immigrant identity as monolithic rather than gender-differentiated. The immigrant aspiration paradox documents high ambitions but does not explain when those ambitions become liabilities. The tracking literature shows that institutional sorting amplifies inequality but does not identify the specific channel through which it does so for immigrant girls. This paper combines all four. The glass ceiling is a gendered aspiration threshold, carried by immigrant families, that activates only when the institutional environment demands a level of academic commitment that conflicts with traditional gender roles. The mechanism is not discrimination, not material deprivation, and not peer rejection. It is the interaction between cultural gender norms and institutional pressure.

3 Hypothesis: the glass ceiling in aspiring cultures

The literature above supplies two independent observations that, taken together, generate a sharp prediction. The first is the *immigrant aspiration paradox*: immigrant families consistently report higher educational aspirations for their children than comparable native families, even after controlling for socioeconomic background (Feliciano and Lanuza, 2017). Children of immigrants attribute greater importance to educational achievement and translate this into higher ambitions. In Germany specifically, immigrant students aim higher than their non-immigrant counterparts despite facing systematic disadvantages at school (Pomianowicz, 2024). The positive immigrant coefficient in Gymnasium (+0.22 SD) is a direct manifestation of this paradox: immigrant boys who reach the academic track outperform native boys, driven by the aspirational premium their families provide.

The second observation is that immigrant families carry traditional gender norms from their countries of origin into the host society, and these norms are remarkably persistent across generations. Parents from achievement-oriented cultures with traditional gender beliefs rate sons' math ability higher and daughters' lower, and these parental beliefs predict children's own self-concept (Nollenberger et al., 2016). Adolescents from immigrant groups in European countries hold more traditional gender beliefs than their majority peers. Crucially, these norms do not suppress girls' aspirations uniformly. They impose a *ceiling* that activates when the institutional environment demands a level of academic identity commitment that conflicts with the family's gender expectations.

The glass ceiling metaphor, borrowed from the labor market literature, captures the mechanism precisely. A glass ceiling is not a barrier at the bottom; it is an invisible constraint that binds only at the top. Immigrant girls in Comprehensive schools, where academic demands are moderate and the peer environment is heterogeneous, do not hit the ceiling. The family's aspiration premium supports both sons and daughters because the level of academic commitment required does not trigger a conflict with traditional gender roles. But in Gymnasium, where sustained, visible academic investment is the price of admission to the

peer group, the ceiling activates. The family continues to invest heavily in the son's academic trajectory, which is culturally congruent with the male provider role, but redirects pressure on the daughter toward community obligations, domestic competence, or at minimum withdraws the active encouragement that the son receives.

This is not a story about immigrant families being unsupportive. It is a story about support being conditional on the institutional demand. When the demand is moderate, immigrant culture is an asset for both genders. When the demand is high, it becomes an asset for boys and a liability for girls. The glass ceiling is unshattered not because immigrant families lack ambition, but because their ambition is gendered in ways that bind only in high-pressure academic environments.

Non-immigrant families, by contrast, do not exhibit this asymmetry. The gender gap among non-immigrants is uniform across tracks because native German families, while subject to their own gender stereotypes, do not carry the same intensity of traditional gender role enforcement. The aspiration premium is smaller, and the ceiling, if it exists, does not differentially activate by track.

This account generates several falsifiable predictions beyond the patterns already documented in the data:

1. In general, immigrants aspire more;

2. Gymnasium immigrant girls aspire less than immigrant girls in the comprehensive track, due to high peer pressure and lack of support;

3. Parental educational aspirations for daughters relative to sons should be lower among immigrant families in Gymnasium than in Comprehensive.

4 Data

4.1 The NEPS

The National Educational Panel Study (NEPS) is a large-scale longitudinal study of education and competence development in Germany. I use two starting cohorts:

- **Starting Cohort 2, SC2:** Students first assessed in Grade 4 in 2008, tracked into secondary school at Grade 5, and followed through Grades 7 and 9. Version 11.0.0. $N = 1{,}311$ in the analysis sample.
- **Starting Cohort 3, SC3:** Students first assessed in Grade 5 in 2010, already in secondary school, and followed through Grades 7 and 9. Version 13.1.0. $N = 2{,}193$ in the analysis sample.

Both cohorts administer standardized math competence tests at Grades 7 and 9 using the same IRT framework with common item anchoring, enabling pooled estimation.

4.2 Outcome variable

The dependent variable is the **corrected Weighted Likelihood Estimate (WLE) of mathematical competence** at Grade 9 (`mag9_sc1`). This is an Item Response Theory (IRT) score: a maximum-likelihood estimate of latent math ability, corrected for measurement error, centered near zero with standard deviation approximately one. The "9" refers to Grade 9. The lagged dependent variable (`mag7_sc1`) is the same measure at Grade 7. Because both use common item anchoring, scores are comparable across waves and cohorts.

Table 1: Variable Definitions

Variable	Definition
Math G7	IRT-scaled math WLE at Grade 7. Controls for prior achievement, encompassing innate ability, family background, primary school quality, and track selection.
Female	= 1 if female. The coefficient of interest: captures the gender gap in G9 *conditional on G7 score*.
Immigrant	= 1 if migration background. SC2: generation status > 0, variable `p400500_g1`. SC3: migration background indicator > 1, variable `p400000_g1`.
High SES	= 1 if household has 101+ books, categories 4–6 on the 6-point ordinal `p34005a`. Standard proxy for parental education and cultural capital in NEPS.
Female × Immigrant	Does the gender gap differ for immigrants?
Female × High SES	Does household SES buffer girls against falling behind?

4.3 Regressors

4.4 Sample

The analysis is restricted to **West Germany**, as the data only allows for this binary regional indicator and the West takes up most of the sample. I retain the **three main tracks**: Gymnasium, Realschule, and Comprehensive. Hauptschule and Förderschule are excluded for small N. Students must have **valid G7 and G9 math scores** and **non-missing SES**. The pooled sample contains $N = 3{,}504$ students.

Table 2: Descriptive Statistics by Track, Pooled SC2 + SC3

	N	% Female	% Immigrant	% High SES	Mean G7	Mean G9	SD G9
Comprehensive	424	54.0	22.9	62.0	−0.20	−0.56	1.11
Realschule	780	45.1	17.3	56.9	0.33	−0.36	0.95
Gymnasium	2,300	52.0	20.3	79.5	1.09	0.59	1.11
Total	3,504						

5 Empirical Strategy

5.1 Value-added specification

The baseline specification, estimated separately within each track j, is:

$$\text{mag9}_{ij} = \alpha_j + \beta_{1j} \cdot \text{mag7}_{ij} + \beta_{2j} \cdot \text{Female}_i + \gamma_j \cdot X_i + \epsilon_{ij} \tag{1}$$

where mag9_{ij} is the math WLE at Grade 9 for student i in track j, mag7_{ij} is the same at Grade 7, and X_i includes immigration status and SES.

5.2 Identification: the diff-in-diff logic

The specification is a difference-in-differences of the gender gap. Consider a boy and a girl who score identically at G7. If $\beta_2 < 0$, the girl scores lower at G9 despite starting at the same level. The first difference is across gender (boy vs. girl); the second difference is across time (G7 vs. G9), absorbed by conditioning on G7. The coefficient β_2 estimates the gender gap that *develops* during G7–G9, net of prior achievement. A negative β_2 means girls fall behind; a positive β_2 means girls catch up.

5.3 Progressive specifications

I estimate four models within each track:

1. **Baseline:** G9 $\sim$ G7 + Female.
2. **Demographics:** + Immigrant + High SES.
3. **Immigration interaction:** + Female $\times$ Immigrant.
4. **Full:** + Female $\times$ High SES.

The stability of the Female coefficient from model 1 to model 2 tests whether the gender gap is an artifact of compositional differences across demographic groups. The interactions in models 3 and 4 test whether the gap is moderated by immigration and SES, the channels that the aspiration literature identifies.

6 Results

Figure 1 provides the descriptive overview. The gender gap, measured as Cohen's d with male minus female, is plotted at G7 and G9 by track and immigration status. The gap widens from G7 to G9 in every subgroup, but the widening is steepest for immigrant students in Gymnasium.

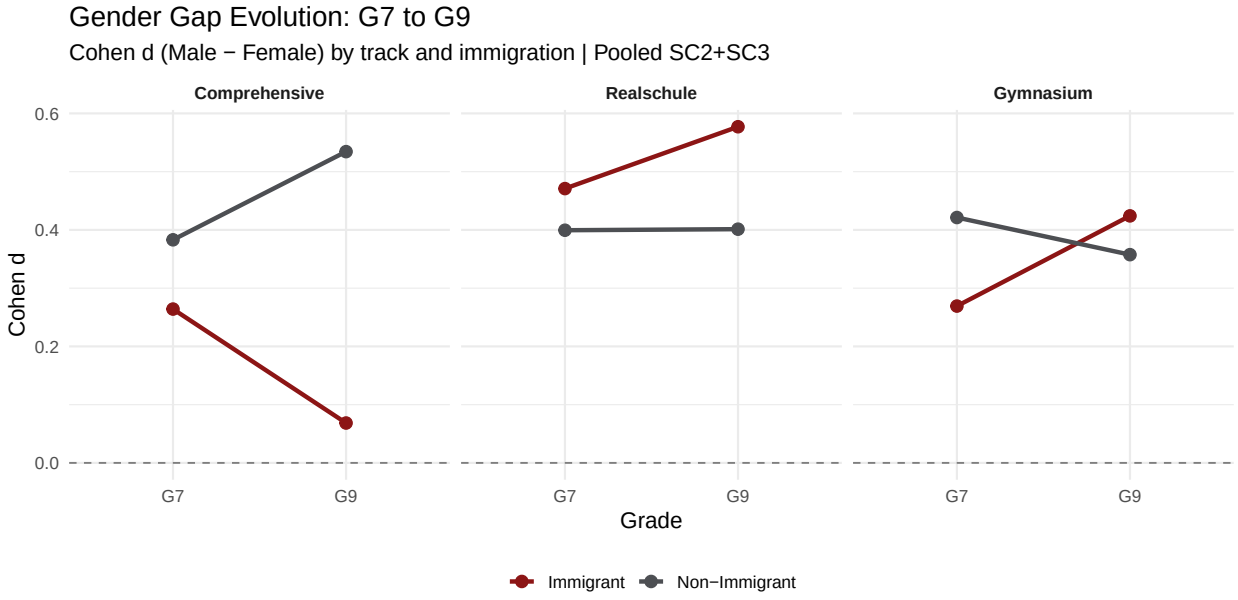

Figure 1: Gender gap evolution from G7 to G9 by track and immigration status. Cohen's d, defined as Male − Female, where positive values indicate male advantage. The gap widens in every track, but immigrant students in Gymnasium show the steepest divergence. Pooled SC2 + SC3, $N = 3{,}504$.

Tables 3–5 report the value-added regressions.

Table 3: Comprehensive Schools ($N = 424$)

	(1)	(2)	(3)	(4)
Math G7	0.568^{***}	0.525^{***}	0.523^{***}	0.522^{***}
	(0.034)	(0.035)	(0.035)	(0.035)
Female	-0.211^{*}	-0.223^{**}	-0.315^{***}	-0.437^{**}
	(0.084)	(0.082)	(0.093)	(0.151)
Immigrant		-0.181	-0.401^{**}	-0.432^{**}
		(0.100)	(0.145)	(0.148)
High SES		0.267^{**}	0.265^{**}	0.170
		(0.090)	(0.089)	(0.130)
Female × Immigrant			0.402^{*}	0.454^{*}
			(0.192)	(0.199)
Female × High SES				0.176
				(0.172)
R^2	0.427	0.447	0.453	0.454
Adj. R^2	0.424	0.442	0.446	0.446
N	424	424	424	424

$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$. Dependent variable: Math WLE at G9. Both cohorts pooled.

6.1 The gender gap widens across all tracks

The Female coefficient is negative and significant in Column 1 of every table: -0.211 in Comprehensive, -0.177 in Realschule, and -0.158 in Gymnasium, all at conventional significance levels. Girls who scored identically to boys at G7 end up between 0.16 and 0.21 standard deviations lower at G9. The coefficient is stable from Columns 1 to 2; adding immigration status and SES barely moves it, confirming that the gap is not a compositional artifact. It develops *within* demographic groups.

6.2 Immigration moderates the gap, but differently by track

The key finding is in Column 3. In **Gymnasium**, as reported in Table 5, the Female × Immigrant interaction is -0.184 and statistically significant. Immigrant boys in Gymnasium actually outperform non-immigrant boys, with a positive and highly significant Immigrant

Table 4: Realschule ($N = 780$)

	(1)	(2)	(3)	(4)
Math G7	0.557^{***}	0.543^{***}	0.543^{***}	0.543^{***}
	(0.027)	(0.027)	(0.027)	(0.027)
Female	-0.177^{**}	-0.177^{**}	-0.157^{**}	-0.190^{*}
	(0.055)	(0.054)	(0.060)	(0.089)
Immigrant		-0.113	-0.057	-0.062
		(0.072)	(0.101)	(0.101)
High SES		0.128^{*}	0.129^{*}	0.104
		(0.054)	(0.054)	(0.073)
Female × Immigrant			-0.112	-0.100
			(0.140)	(0.142)
Female × High SES				0.054
				(0.109)
R^2	0.388	0.395	0.395	0.396
Adj. R^2	0.386	0.392	0.392	0.391
N	780	780	780	780

$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$. Dependent variable: Math WLE at G9. Both cohorts pooled.

main effect of +0.223, consistent with positive selection into the academic track. But immigrant girls do not share this advantage: the interaction wipes out the immigrant premium for girls and adds to the gender gap. The total Female effect for immigrant girls in Gymnasium is $-0.121 + (-0.184) = -0.305$, roughly double the gap for non-immigrant girls. This is the "caught between cultures" dynamic: the high-aspiration academic environment amplifies identity conflict for immigrant girls.

In **Comprehensive schools**, as reported in Table 3, the pattern reverses. The Female × Immigrant interaction is +0.402 and statistically significant. Immigrant girls are relatively *protected* from the gender gap that non-immigrant girls experience. The total Female effect for immigrant girls is $-0.315 + 0.402 = +0.087$, effectively zero, or even slightly positive. The lower-pressure Comprehensive environment does not generate the same identity conflict. The Immigrant main effect is -0.401, so immigrants generally perform worse in Comprehensive, but the gender gap within immigrants is near zero.

Table 5: Gymnasium (N = 2,300)

	(1)	(2)	(3)	(4)
Math G7	0.572^{***}	0.571^{***}	0.572^{***}	0.572^{***}
	(0.016)	(0.017)	(0.017)	(0.017)
Female	-0.158^{***}	-0.159^{***}	-0.121^{**}	-0.183^{*}
	(0.037)	(0.037)	(0.042)	(0.086)
Immigrant		0.129^{**}	0.223^{***}	0.218^{***}
		(0.046)	(0.065)	(0.066)
High SES		0.209^{***}	0.206^{***}	0.168^{**}
		(0.046)	(0.046)	(0.065)
Female × Immigrant			-0.184^{*}	-0.173
			(0.091)	(0.091)
Female × High SES				0.075
				(0.091)
R^2	0.369	0.376	0.377	0.378
Adj. R^2	0.369	0.375	0.376	0.376
N	2,300	2,300	2,300	2,300

$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$. Dependent variable: Math WLE at G9. Both cohorts pooled.

In **Realschule**, as reported in Table 4, the interaction is small and insignificant at -0.112. The intermediate track produces a uniform gender gap regardless of immigration status.

6.3 SES does not independently moderate the gap

The Female × High SES interaction is positive but insignificant in all three tracks, ranging from 0.054 in Realschule to 0.176 in Comprehensive. Household cultural capital, as proxied by books, does not significantly buffer girls against the widening gap once immigration is controlled. The action is in immigration status, not SES.

6.4 Occupational aspirations: immigrant premium is real

If the glass ceiling operates through aspirations, it should show up not only in test scores but in what students want to become. I regress realistic occupational aspiration, measured by

the SIOPS-88 prestige scale ranging from 13 to 78, on G7 math, gender, immigration, SES, and the Female × Immigrant interaction, separately by track. Table 6 reports the results.

Table 6: Occupational Aspiration by Track, SIOPS Prestige

	Comprehensive			Realschule			Gymnasium		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Math G7	2.42***	2.91***	2.86***	0.32	0.68	0.66	1.22***	1.39***	1.39***
	(0.73)	(0.75)	(0.75)	(0.54)	(0.54)	(0.55)	(0.30)	(0.30)	(0.30)
Female	4.12*	4.14*	3.10	4.69***	4.63***	4.36***	2.43***	2.53***	2.34**
	(1.83)	(1.79)	(2.02)	(1.07)	(1.05)	(1.15)	(0.69)	(0.68)	(0.77)
Immigrant		8.49***	5.66		5.60***	4.76*		4.19***	3.72**
		(2.18)	(3.38)		(1.44)	(2.06)		(0.83)	(1.19)
High SES		0.62	0.67		0.64	0.65		2.37**	2.38**
		(1.96)	(1.96)		(1.06)	(1.06)		(0.81)	(0.81)
Female × Immig			4.69			1.62			0.90
			(4.26)			(2.82)			(1.63)
R^2	0.051	0.105	0.109	0.036	0.064	0.065	0.017	0.037	0.037
N	264	264	264	517	517	517	1,446	1,446	1,446

$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$. DV: SIOPS-88 occupational prestige, scale 13–78. SC2 wave 9, SC3 wave 5. Pooled.

Two patterns are noteworthy. First, the **immigrant aspiration premium is real**: the Immigrant main effect is positive and highly significant in all three tracks in Column 2, ranging from +4.19 in Gymnasium to +8.49 in Comprehensive. Immigrant students aim for more prestigious occupations than natives, conditional on math ability. This confirms the immigrant aspiration paradox documented by Feliciano and Lanuza (2017): immigrant families transmit higher ambitions even when their children face structural disadvantages.

Second, the Female × Immigrant interaction does not reach significance in any track, though the coefficient magnitude is informative. In Comprehensive the interaction is +4.69, rather positive though highly variant. Immigrant girls, on average, aim higher than immigrant boys. In Gymnasium it is near zero at +0.90: the immigrant aspiration premium is equally shared between genders. The diminished motivation among girls in the gymnasium is reflected in their academic skill development.

7 Discussion

The results are consistent with the glass ceiling hypothesis laid out in Section 3. Immigrant families carry an aspiration premium that benefits their children, but the premium is gendered, and this gendered differentiation activates only when the institutional environment demands sustained, high-intensity academic commitment. In Gymnasium, that demand triggers the ceiling: the family's traditional gender norms redirect investment away from daughters and toward sons, not out of malice but out of a culturally inherited understanding of what constitutes appropriate ambition for each gender. In Comprehensive schools, the demand is lower, the ceiling does not bind, and immigrant girls benefit from the same cultural emphasis on education that immigrant boys do.

The occupational aspiration result clinches the aspirational interpretation. The glass ceiling shows up not just in math scores but in SIOPS occupational prestige. If the math gap were driven solely by merit and curriculum difficulty, it would not extend to self-reported aspirations. The fact that it does means the mechanism runs through the student's own expectation formation, precisely the value-distortion channel that Penn's model predicts.

The pattern also maps onto Penn's threshold model. In Penn's framework, a class-specific aspiration threshold s^* determines whether interpersonal comparison motivates or demobilizes. The glass ceiling can be understood as a *gendered threshold*: for immigrant boys, the threshold is set by academic peers and is attainable through effort. For immigrant girls, the threshold is set by academic peers *and* constrained by family expectations, making it higher in effective terms. The result is that immigrant girls in Gymnasium are more likely to fall below the effective threshold and enter Penn's lock-in regime: not because they lack ability, but because the combination of high academic demand and traditional gender expectations makes the aspiration gap too large to bridge through effort alone.

The null result on SES reinforces the cultural interpretation. If the mechanism were material, if high-SES families simply invested more in daughters, then Female $\times$ High SES should be positive and significant. It is not. The glass ceiling is not about money or books

in the household. It is about the cultural expectations that immigrant families bring, and how those expectations interact with the institutional pressure of the academic track.

An important caveat is that the current analysis cannot distinguish between family-side mechanisms, such as differential parental investment, and community-side mechanisms, such as peer pressure within immigrant social networks, in a causal way. Nonetheless, in the following analysis we might find complementary evidence that compensates for lack of identification strategies.

8 Matched sample and quasi-experimental evidence

The value-added regressions condition on G7 scores, but students are already sorted by G7. A sharper test uses pre-tracking scores. Using the SC2 cohort, I match students on G4 math scores across tracks within the Q10–Q90 common support, trimming the Gymnasium upper tail. The matched sample contains 626 students with balanced G4 means: $F(2, 623) = 0.151$, $p = 0.860$.

Figure 2 shows the divergence: students who scored identically at G4 end up 0.57 SD apart by G9, depending on track assignment. This seems like a prima facie treatment effect of tracking on math growth. What it does show is that the comprehensive average is on a par with the real school average. Parents who opt out of the tracking system are usually recommended for a real school instead of a gymnasium. Therefore, it is reasonable to conjecture that parents perceive different tracks as having different effects on student performances.

Figure 3 reports the Female and Female $\times$ Immigrant coefficients from the value-added regression on the matched sample. The Gymnasium Female coefficient *doubles* to -0.355, highly significant, relative to the pooled estimate, confirming that the pooled sample understates the gap. The Female $\times$ Immigrant interaction in matched Gymnasium is -0.280, consistent with the glass ceiling hypothesis. Moreover, the comprehensive coefficient is pos-

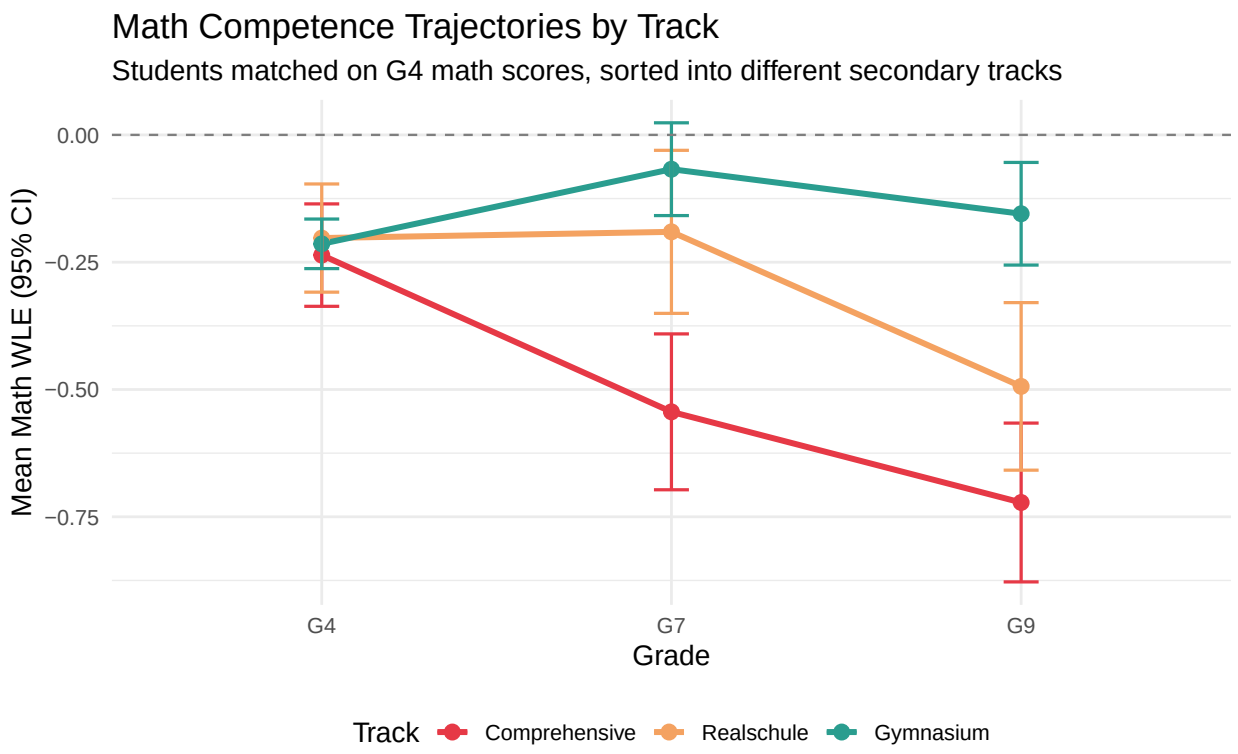

Figure 2: Math competence trajectories by track in the matched sample. Students are balanced on G4 math scores by construction. Gymnasium students maintain or improve; Comprehensive students lose ≈ 0.5 SD by G9. SC2 only, $N = 626$.

itive, albeit with noise and lack of power. Tables 7–9 provide the full regression output.

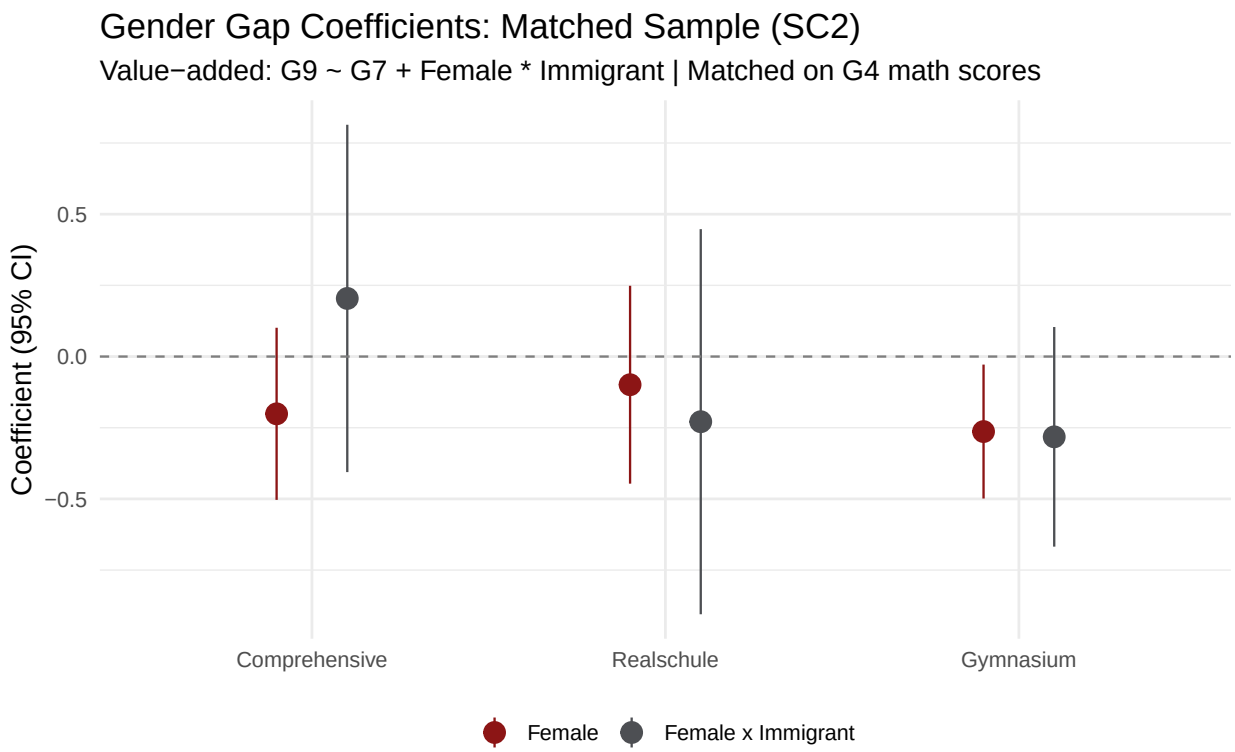

Figure 3: Value-added coefficients on Female and Female × Immigrant in the matched sample, SC2. Reference: male non-immigrant. The Gymnasium Female coefficient of −0.36 is more than double the pooled estimate. Error bars: 95% CI.

9 Structural Estimation: Penn's Threshold in Comprehensive vs. Gymnasium

Penn (2017) predicts that a class-specific aspiration threshold $\bar{s}$ divides students into two regimes: above it, interpersonal comparison motivates effort and scores converge with low persistence γ_{above}; below it, the status gap discourages investment and scores lock in with

Table 7: Comprehensive Schools, Matched Sample ($N = 180$)

	(1)	(2)	(3)	(4)
Math G7	0.577^{***}	0.558^{***}	0.553^{***}	0.554^{***}
	(0.064)	(0.064)	(0.064)	(0.065)
Female	−0.154	−0.143	−0.190	−0.149
	(0.133)	(0.132)	(0.152)	(0.235)
Immigrant		−0.183	−0.287	−0.279
		(0.154)	(0.227)	(0.230)
High SES		0.245	0.243	0.276
		(0.134)	(0.134)	(0.197)
Female × Immigrant			0.191	0.180
			(0.307)	(0.312)
Female × High SES				−0.061
				(0.269)
R^2	0.346	0.366	0.367	0.367
N	180	180	180	180

$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$. Matched on G4 math (SC2 only). DV: Math WLE G9.

Table 8: Realschule, Matched Sample ($N = 130$)

	(1)	(2)	(3)	(4)
Math G7	0.476^{***}	0.473^{***}	0.468^{***}	0.468^{***}
	(0.081)	(0.080)	(0.080)	(0.080)
Female	−0.166	−0.141	−0.083	−0.262
	(0.150)	(0.150)	(0.173)	(0.275)
Immigrant		−0.063	0.049	0.038
		(0.168)	(0.239)	(0.239)
High SES		0.323^{*}	0.323^{*}	0.198
		(0.156)	(0.156)	(0.216)
Female × Immigrant			−0.224	−0.199
			(0.337)	(0.339)
Female × High SES				0.263
				(0.313)
R^2	0.233	0.261	0.263	0.267
N	130	130	130	130

$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$. Matched on G4 math (SC2 only). DV: Math WLE G9.

Table 9: Gymnasium, Matched Sample ($N = 316$)

	(1)	(2)	(3)	(4)
Math G7	0.404^{***}	0.394^{***}	0.397^{***}	0.399^{***}
	(0.057)	(0.058)	(0.058)	(0.058)
Female	-0.355^{***}	-0.361^{***}	-0.259^{*}	-0.330
	(0.095)	(0.096)	(0.120)	(0.234)
Immigrant		-0.077	0.069	0.065
		(0.098)	(0.142)	(0.142)
High SES		0.137	0.135	0.082
		(0.119)	(0.118)	(0.191)
Female × Immigrant			-0.280	-0.274
			(0.196)	(0.197)
Female × High SES				0.085
				(0.241)
R^2	0.195	0.200	0.205	0.206
N	316	316	316	316

$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$. Matched on G4 math (SC2 only). DV: Math WLE G9.

high persistence γ_{below}. The reduced form is a piecewise linear model:

$$s^i_{t+1} = \begin{cases} a + \gamma_1 \cdot s^i_t + \epsilon_i & \text{if } s^i_t > \bar{s} \\ a + \gamma_2 \cdot s^i_t + \epsilon_i & \text{if } s^i_t \leq \bar{s} \end{cases} \tag{2}$$

with Penn's prediction $\gamma_1 < \gamma_2$. I estimate $\bar{s}$ by Hansen (2000) grid search on the full unmatched pooled sample for the G7→G9 transition, separately by track, gender, and immigration status. This produces a 2×4 typology: two tracks, Comprehensive and Gymnasium, crossed with four demographic cells of Native Male, Immigrant Male, Native Female, and Immigrant Female. Figure 4 reports all eight cells.

The eight panels tell a clean story. In seven of them, the pattern is the same: persistence is higher above the threshold than below, or the two regimes are nearly symmetric. Students who clear the aspiration cutoff see their effort compound; those below it are mobile, not trapped. This holds for native boys and native girls in both tracks, for immigrant boys in

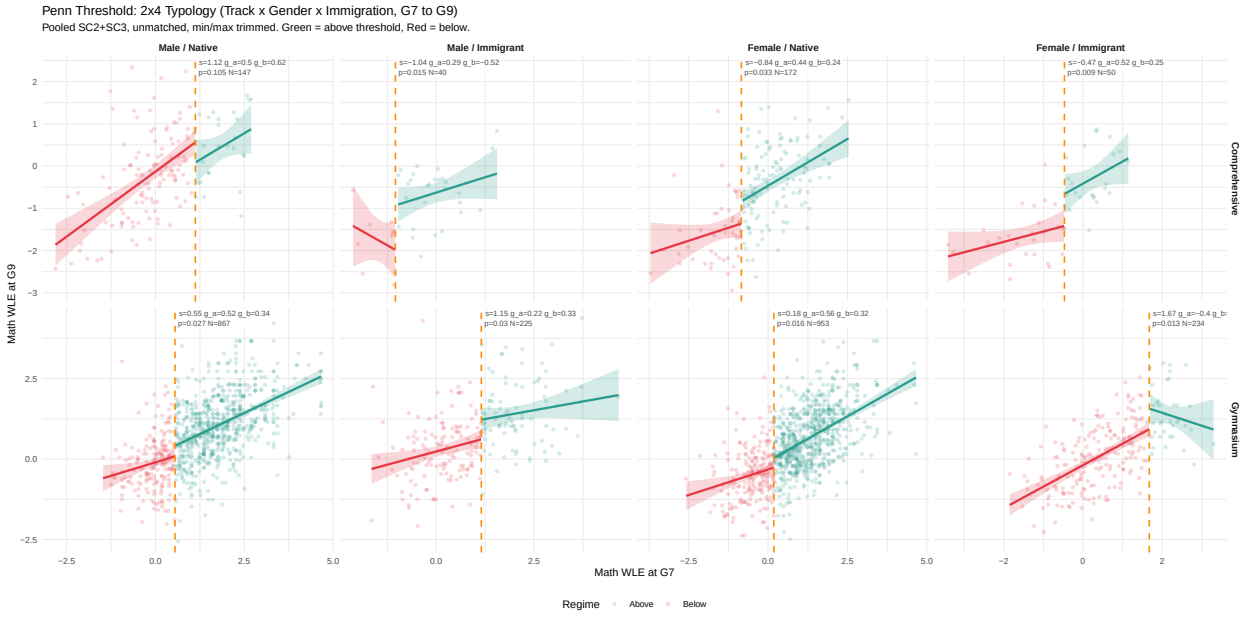

Figure 4: Penn threshold estimation: 2×4 typology of Track $\times$ Gender $\times$ Immigration, G7→G9. Pooled SC2+SC3, unmatched, min/max trimmed. Each panel shows the piecewise regression with the estimated threshold in orange dashed. Green = above threshold; red = below. The aspiration trap, Penn's lock-in where the red slope is steeper than the green, appears in one cell only: **Gymnasium Immigrant Female**.

both tracks, and for immigrant girls in Comprehensive. The aspiration mechanism works as expected: the threshold marks where effort starts to pay off.

One cell breaks the pattern. **Gymnasium Immigrant Girls**, with $N = 234$, are the only group that exhibits Penn's aspiration trap, and they exhibit it in its most extreme form: $\gamma_{\text{above}} = -0.402$, $\gamma_{\text{below}} = 0.668$, $F = 4.46$, $p = 0.013$. Below the threshold at $\bar{s} = 1.67$, scores lock in hard; a girl who falls behind at G7 stays behind at G9. Above the threshold, persistence is not just low but *negative*: the few immigrant girls who clear the threshold actually *regress*. Effort above the cutoff does not compound; it dissipates. This is the structural signature of the glass ceiling. It is not a uniform drag on performance. It is a regime switch in which the normal reward to effort, the pattern that operates for everyone else, is reversed for immigrant girls in the academic track.

The contrast with **Comprehensive Immigrant Girls** is instructive despite the small sample of $N = 50$. Their piecewise structure is significant at $F = 5.18$, $p = 0.009$, and

goes in the opposite direction: $\gamma_{\text{above}} = 0.524$, $\gamma_{\text{below}} = 0.246$. Effort compounds above the threshold. No lock-in below. Immigrant girls in the lower-pressure track experience the same healthy aspiration dynamics as everyone else.

Gymnasium Native Girls, with $N = 953$, confirm that this is not a gender effect per se. Their pattern is $\gamma_{\text{above}} = 0.559$, $\gamma_{\text{below}} = 0.318$ at $p = 0.016$; effort compounds above, mobility below. This is indistinguishable from native boys. The aspiration trap requires the intersection of immigrant status and the academic track. Neither alone is sufficient.

The structural estimation localizes the glass ceiling with precision. It is not girls in general who face the aspiration trap. It is not immigrants in general. It is not the Gymnasium environment in general. It is the specific combination of traditional gender norms, carried by immigrant families, and high-pressure academic demands, imposed by the Gymnasium environment, that produces a regime switch in which effort above the threshold is punished rather than rewarded, and position below it is frozen rather than fluid. The glass ceiling is a piecewise discontinuity in the persistence function, operating at the intersection of gender, immigration, and institutional context.

10 Conclusion

This paper has documented a glass ceiling in the math achievement of immigrant girls in the German tracking system and identified the aspiration mechanism that produces it. The math gender gap widens significantly across all three tracks during secondary school, with girls who started at the same level as boys falling meaningfully behind. The central finding is that immigration status moderates this gap in opposite directions depending on the track. In Gymnasium, immigrant girls face a compounded disadvantage; the high-aspiration academic environment amplifies the gender gap rather than mitigating it. In Comprehensive schools, the pattern reverses: immigrant girls are relatively protected, and the gender gap within immigrants is effectively zero. Structural estimation of Penn's threshold model reveals the

sharpest version of this result. Out of eight demographic cells, Gymnasium immigrant girls are the *only* group that exhibits the aspiration trap, where effort above the threshold is punished rather than rewarded and position below it is frozen rather than fluid. Socioeconomic status, meanwhile, does not moderate the gap in any track, confirming that the mechanism is cultural rather than material.

The glass ceiling hypothesis integrates insights from four previously separate literatures. From the aspiration literature (Ray, 2006; Genicot and Ray, 2017), it takes the idea that effort is non-monotonic in the aspiration gap. From the identity literature (Akerlof and Kranton, 2002; Shayo, 2009), it takes the insight that group membership constrains behavioral prescriptions. From the immigrant aspiration paradox (Kao and Tienda, 1995; Feliciano and Lanuza, 2017; Salikutluk, 2016), it takes the empirical regularity that immigrant families carry higher ambitions. From the tracking literature (Hanushek and Woessmann, 2006; Van de Werfhorst and Mijs, 2010), it takes the institutional mechanism that concentrates peer pressure and curricular demand. The synthesis is the *gendered aspiration threshold*: immigrant families set high aspirations for all children, but enforce a ceiling on daughters that activates only when the institutional environment demands sustained, visible academic identity commitment. The threshold is not biological, not material, and not imposed by the host society. It is carried within the immigrant family's cultural expectations and triggered by institutional pressure.

This framework resolves a puzzle in the tracking literature. Dustmann et al. (2016) find zero average long-term effects of tracking, but this null masks compositional heterogeneity. Tracking simultaneously elevates immigrant boys, who benefit from the aspiration premium in a high-demand environment, and depresses immigrant girls, who hit the gendered ceiling in the same environment. The aggregate effect washes out, but the distributional consequences are large and policy-relevant. The glass ceiling also extends the "caught between cultures" finding of Dahl et al. (2022) by providing a structural mechanism. It is not simply that opportunities backfire for immigrant girls, but that the *level of institutional demand*

determines whether the family's cultural resources are an asset or a liability.

The findings carry direct implications for education policy in tracked systems. Comprehensive schooling appears to protect immigrant girls from the gendered aspiration trap. This does not mean that Gymnasium is harmful in itself; it is the interaction of Gymnasium-level demands with traditional gender norms that produces the ceiling. Policies that reduce the salience of the identity conflict, such as mixed-ability classrooms within Gymnasium, mentoring programs that target immigrant girls specifically, or explicit norm-setting that validates academic ambition for girls, may deactivate the ceiling without dismantling the track. The occupational aspiration results suggest that intervention at the aspiration level, not just the skill level, is necessary. Programs like the aspiration workshops documented by Orkin et al. (2023) and the female leadership exposure studied by Beaman et al. (2012) may be more effective for immigrant girls than tutoring or curricular enrichment, because the binding constraint is not ability but the cultural permission to aspire. The null result on SES underscores that the glass ceiling is not a poverty story. Household cultural capital does not buffer immigrant girls. Policies targeting material disadvantage, such as scholarships, free textbooks, and after-school programs, will not on their own close the gap. The mechanism runs through cultural expectations, not resource constraints, and policy must engage the family's understanding of gendered ambition if it is to be effective.

Several limitations warrant caution. The analysis cannot distinguish between family-side mechanisms, such as differential parental investment in sons versus daughters, and community-side mechanisms, such as peer pressure within immigrant social networks. Both are consistent with the data, and disentangling them requires within-family variation through sibling fixed effects with opposite-sex pairs, which the current NEPS samples do not support with adequate power. "Immigrant" is also a coarse category. The NEPS migration background indicator pools Turkish, Eastern European, Middle Eastern, and other origin groups that differ substantially in gender norms. The glass ceiling may be stronger for some origin groups than others, and a finer decomposition would sharpen the mechanism. The

matched sample analysis relies on one cohort only, limiting statistical power in the interaction terms. The matched Gymnasium interaction coefficient is substantively large but does not reach conventional significance, consistent with the hypothesis but not dispositive. The Penn threshold estimation in the Comprehensive Immigrant Female cell rests on a small number of observations; while the point estimates are instructive, replication with larger samples is essential.

Future work should pursue three extensions. The mechanism can be tested directly by measuring parental aspirations for sons versus daughters within immigrant families across tracks, a prediction that the hypothesis generates and that NEPS parental surveys can address. Quasi-experimental variation in track assignment, from state-level differences in tracking age, teacher recommendation policies, or randomized mentoring interventions, can strengthen causal identification. The framework should also be tested outside Germany. Any tracked system with a substantial immigrant population and origin-country variation in gender norms, such as Austria, the Netherlands, or Switzerland, should exhibit the same glass ceiling pattern, because the mechanism depends on the interaction of institutional pressure and cultural gender expectations, not on features specific to the German context.

The glass ceiling is unshattered not because immigrant families lack ambition, but because their ambition is gendered in ways that bind only in high-pressure academic environments. The aspiration premium that immigrant families carry is real, powerful, and beneficial, but it is not gender-neutral, and the tracking system amplifies its asymmetry. Understanding this interaction is a prerequisite for designing institutions that serve all students equally.

References

Akerlof, G. A. and Kranton, R. E. (2002). Identity and schooling: Some lessons for the economics of education. *Journal of Economic Literature*, 40(4):1167–1201.

Austen-Smith, D. and Fryer Jr, R. G. (2005). An economic analysis of “acting white”. *The*

Quarterly Journal of Economics, 120(2):551–583.

Beaman, L., Duflo, E., Pande, R., and Topalova, P. (2012). Female leadership raises aspirations and educational attainment for girls: A policy experiment in india. *Science*, 335(6068):582–586.

Becker, B., Gresch, C., and Zimmermann, T. (2023). Are they still aiming high? the development of educational aspirations of lower secondary school students with immigrant backgrounds in germany. *International Migration Review*, 57(2).

Bisin, A., Patacchini, E., Verdier, T., and Zenou, Y. (2011). Formation and persistence of oppositional identities. *European Economic Review*, 55(8):1046–1071.

Brunello, G. and Checchi, D. (2007). Does school tracking affect equality of opportunity? new international evidence. *Economic Policy*, 22(52):782–861.

Carlana, M. (2019). Implicit stereotypes: Evidence from teachers' gender bias. *Quarterly Journal of Economics*, 134(3):1163–1224.

Carlana, M., La Ferrara, E., and Pinotti, P. (2022). Goals and gaps: Educational careers of immigrant children. *Econometrica*, 90(1):1–29.

Crul, M. and Schneider, J. (2010). Comparative integration context theory: Participation and belonging in new diverse european cities. *Ethnic and Racial Studies*, 33(7):1249–1268.

Dahl, G. B., Felfe, C., Frijters, P., and Rainer, H. (2022). Caught between cultures: Unintended consequences of improving opportunity for immigrant girls. *Review of Economic Studies*, 89(5):2491–2528.

Dalton, P. S., Ghosal, S., and Mani, A. (2016). Poverty and aspirations failure. *The Economic Journal*, 126(590):165–188.

Dollmann, J. (2021). Ethnic inequality in choice- and performance-driven education systems: A longitudinal study of educational choices in england, germany, the netherlands, and sweden. *British Journal of Sociology*, 72(5):1345–1364.

Dustmann, C., Puhani, P. A., and Schönberg, U. (2016). The long-term effects of early track choice. *The Economic Journal*, 126(594):1348–1380.

Feliciano, C. and Lanuza, Y. R. (2017). An immigrant paradox? contextual attainment and intergenerational educational mobility. *American Sociological Review*, 82(1):211–241.

Genicot, G. and Ray, D. (2017). Aspirations and inequality. *Econometrica*, 85(2):489–519.

Guiso, L., Monte, F., Sapienza, P., and Zingales, L. (2008). Culture, gender, and math. *Science*, 320(5880):1164–1165.

Hansen, B. E. (2000). Sample splitting and threshold estimation. *Econometrica*, 68(3):575–603.

Hanushek, E. A. and Woessmann, L. (2006). Does educational tracking affect performance and inequality? differences-in-differences evidence across countries. *The Economic Journal*, 116(510):C63–C76.

Jürges, H. and Schneider, K. (2011). Why young boys stumble: Early tracking, age and gender bias in the german school system. *German Economic Review*, 12(4):371–394.

Kao, G. and Tienda, M. (1995). Optimism and achievement: The educational performance of immigrant youth. *Social Science Quarterly*, 76(1):1–19.

Lüdemann, E. and Schwerdt, G. (2013). Migration background and educational tracking: Is there a double disadvantage for second-generation immigrants? *Journal of Population Economics*, 26(2):455–481.

Nollenberger, N., Rodríguez-Planas, N., and Sevilla, A. (2016). The math gender gap: The role of culture. *American Economic Review*, 106(5):257–261.

Orkin, K., Garlick, R., Mahmud, M., Sedlmayr, R., Haushofer, J., and Dercon, S. (2023). Aspiring to a better future: Can a simple psychological intervention reduce poverty? NBER Working Paper 31735.

Penn, E. M. (2017). Inequality, social context, and value divergence. *The Journal of Politics*, 79(1):153–165.

Phalet, K. and Schönpflug, C. (2001). Intergenerational transmission in turkish immigrant families: Parental collectivism, achievement values and gender differences. *Journal of Comparative Family Studies*, 32(4):489–504.

Pomianowicz, K. (2024). Educational expectation gaps between second-generation immigrant and ethnic majority students in a comparative perspective: The moderating role of educational tracking. *International Migration Review*, 58(1):275–310.

Ray, D. (2006). Aspirations, poverty, and economic change. *Understanding Poverty*, pages 409–443. In Banerjee, Benabou, and Mookherjee (eds.), Oxford University Press.

Roland G. Fryer, J. and Torelli, P. (2010). An empirical analysis of 'acting white'. *Journal of Public Economics*, 94(5–6):380–396.

Ruhose, J. and Schwerdt, G. (2016). Does early educational tracking increase migrant–native achievement gaps? differences-in-differences evidence across countries. *Economics of Education Review*, 52:134–154.

Salikutluk, Z. (2016). Why do immigrant students aim high? explaining the aspiration–achievement paradox of immigrants in germany. *European Sociological Review*, 32(5):581–592.

Shayo, M. (2009). A model of social identity with an application to political economy: Nation, class, and redistribution. *American Political Science Review*, 103(2):147–174.

Stoet, G. and Geary, D. C. (2018). The gender-equality paradox in science, technology, engineering, and mathematics education. *Psychological Science*, 29(4):581–593.

Tjaden, J. D. and Hunkler, C. (2017). The optimism trap: Immigrants and educational choices in stratified education systems. *Social Science Research*, 67:213–228.

Tobisch, A. and Dresel, M. (2017). Negatively or positively biased? dependencies of teachers' judgments and expectations based on students' ethnic and social backgrounds. *Social Psychology of Education*, 20(3):731–752.

Van de Werfhorst, H. G. and Mijs, J. J. B. (2010). Achievement inequality and the institutional structure of educational systems: A comparative perspective. *Annual Review of Sociology*, 36:407–428.